SlopeGuard AI — Meghalaya Landslide Risk Monitor

Smart India Hackathon 2026 Prototype Documentation | SIH26001: AI-Based Early Warning System

Prototype Disclosure: SlopeGuard AI is a decision-support prototype built on official Meghalaya Government GIS datasets, real-time Open-Meteo rainfall telemetry, and a multi-factor weighted risk engine. It provides explainable risk scores (0-100) and actionable response guidance for regional disaster planners.

1. Project Overview

SlopeGuard AI is an early warning and spatial intelligence decision-support platform designed for landslide risk monitoring across the 12 districts of Meghalaya, North East India. Meghalaya's steep topographic relief combined with intense monsoon precipitation causes severe recurring slope failures along key corridors such as NH-6. SlopeGuard AI aggregates spatial GIS data, real-time rainfall forecasts, and nearby infrastructure layers to help authorities evaluate risk, coordinate inspections, and collect ground reports.

2. Problem Statement (SIH26001)

Problem Statement SIH26001 focuses on 'AI-Based Early Warning and Landslide Risk Monitoring System in the North Eastern Region'. North Eastern states face severe socio-economic disruptions due to monsoonal landslides cutting off lifeline corridors.

- Limited integrated platforms that combine official government spatial data with live meteorological data.
- Black-box predictive algorithms that lack interpretability and transparent factor breakdowns for field officers.
- Absence of rapid critical infrastructure exposure evaluation (highways, villages, healthcare, education).
- Lack of resilient citizen reporting tools that function across low-bandwidth or offline rural environments.

3. Proposed Solution

SlopeGuard AI addresses these challenges through a modular, transparent, full-stack architecture that combines official government GIS boundaries and 1,733 historical landslide records with real-time digital elevation and rainfall models. The platform offers interactive spatial exploration, multi-factor risk indexing, automated infrastructure exposure detection, and offline-first citizen incident reporting.

4. Key Implemented Features

- High-Performance GIS Dashboard: Interactive Leaflet map rendering all 12 Meghalaya district boundaries and 1,733 historical landslide markers using GPU canvas acceleration.
- Multi-Factor Risk Engine: Computes a 0–100 risk score and categorizes risk into LOW, MODERATE, HIGH, or CRITICAL with plain-language factor breakdown.
- Dynamic Scenario Simulation: Real-time slider to simulate normal (25mm), heavy (75mm), and extreme (200mm+) monsoon downpours across statewide grid cells.
- Infrastructure Exposure Analysis: Live OpenStreetMap Overpass queries detecting nearby roads, villages, schools, hospitals, and bridges within 1-5 km.
- Demonstration Alerts & Response Workflow: Automated warning generation for High/Critical zones with lifecycle tracking (Active -> Acknowledged -> Resolved).
- Field & Citizen Verification Reports: Community ground reporting with offline queue in localStorage for seamless sync upon reconnection.

5. System Architecture

SlopeGuard AI is implemented as a unified TypeScript full-stack application using TanStack Start and Nitro server routes on Vite. The architecture separates client mapping components from server-only GIS geometry operations to guarantee high client performance and security.

- Client Layer: React 19 single-page application with Leaflet dynamic map loaders and TanStack Query state caching.
- Server API Layer: Nitro-powered TypeScript server routes (/api/districts, /api/landslides, /api/risk, /api/alerts, /api/reports).
- GIS Processing: Server-side spatial calculations, Douglas-Peucker simplified polygons, and Haversine proximity joins.
- Persistence: Supabase PostgreSQL cloud database for alert lifecycles and citizen reports, with built-in client fallback.

6. Technology Stack

Frontend & Mapping

- React 19 & TypeScript: Declarative, strictly typed UI layer.
- Vite 8 & Tailwind CSS v4: High-performance build tool and modern tokenized styling.
- Leaflet 1.9 & React-Leaflet 5: Interactive GIS mapping with canvas-rendered vector layers.
- TanStack Router & Query v5: Type-safe client routing and aggressive data caching (1h TTL).
- Radix UI & Lucide Icons: Accessible interface primitives and iconography.

Backend & Services

- TanStack Start & Nitro: Full-stack SSR and unified TypeScript server endpoints.
- Supabase (@supabase/supabase-js): PostgreSQL database for alerts and report records.
- Zod: Strict runtime schema validation for all API inputs and mutations.
- Open-Meteo API: Live precipitation and meteorological forecasts.
- OpenStreetMap Overpass API: Live spatial queries for surrounding civil infrastructure.

7. Multi-Factor Risk Scoring Engine

The risk engine uses a transparent, weighted formula to calculate the composite landslide risk score (0–100):

- Rainfall Factor (40%): Live precipitation or scenario slider (Normal: 25mm, Heavy: 75mm, Extreme: 200mm+).
- Slope & Relief Factor (25%): Digital elevation gradient derived from regional topographic data.
- Historical Incident Density (25%): Spatial clustering and frequency of past landslides from official MBDA records.
- Infrastructure Exposure (10%): Proximity to primary highways, settlement zones, schools, and health centres.

Risk Tiers: LOW (0–39, routine monitoring), MODERATE (40–69, heightened surveillance), HIGH (70–84, alert eligible / infrastructure inspection), CRITICAL (85–100, urgent response / emergency preparedness).

8. User & System Workflow

- 1. Explore Statewide Risk: User opens dashboard; interactive map renders district polygons and risk-colored grid cells.
- 2. Select District / Point: User clicks a district or map coordinate to view local historical records and terrain parameters.
- 3. Simulate Weather Scenarios: Adjusting the rainfall slider dynamically updates risk scores and affected cell counts.
- 4. Analyze Vulnerability: System queries Overpass API and displays nearby civil infrastructure assets (roads, clinics, schools).
- 5. Issue & Manage Alerts: Eligible high-risk assessments generate demonstration alerts tracked on the /alerts dashboard.
- 6. Submit Ground Reports: Field workers submit ground truth observations (cracks, debris) via /reports, stored offline if disconnected.

9. Project Structure

src/components/: UI components categorized by feature (map, risk, decision, reports, layers, ui).

src/data/: GeoJSON datasets for Meghalaya districts (meghalaya-districts.json) and landslide records (meghalaya-landslides.json).

src/lib/: Core business logic including GIS service, risk model, Overpass proximity queries, and recommendation rules.

src/routes/: File-based TanStack routes including __root.tsx, index.tsx, alerts/, reports.tsx, and api/ endpoints.

supabase/: Database schema definitions and SQL migrations for alerts and reports persistence.

10. Local Setup & Execution Commands

Prerequisites: Node.js v20+ or v24+ and npm v10+ or v11+.

- Install Dependencies: npm install
- Start Development Server: npm run dev (Runs on http://localhost:8080)
- Production Build: npm run build
- Preview Production Build: npm run preview

11. Environment Configuration & Security

SlopeGuard AI is configured with safe local defaults and standalone fallbacks. Cloud database connectivity uses the following environment variables (configured via .env / .env.example):

- SUPABASE_PROJECT_ID: Supabase project identifier.
- SUPABASE_PUBLISHABLE_KEY / VITE_SUPABASE_PUBLISHABLE_KEY: Public client API key.
- SUPABASE_URL / VITE_SUPABASE_URL: Supabase API gateway endpoint.

Security Compliance: All environment files containing secrets are strictly excluded via .gitignore. API write endpoints enforce Zod input validation and sanitize data before storage.

12. Testing & Validation Results

- Application Startup: PASS (Vite dev server initialized on port 8080 with 0 startup errors).
- GIS Rendering: PASS (12 district polygons and 1,733 landslide points loaded and rendered on Leaflet canvas).
- Interactive End-to-End Workflow: PASS (Tested rainfall simulation, risk score calculation, alert lifecycle, and offline report queue).
- Production Build: PASS (vite build and Nitro bundle compilation completed with 0 errors).

13. Future Roadmap

- 1. Multi-spectral Satellite Ingestion: Integrate ESA Sentinel-2 and SAR imagery for automated slope deformation tracking.
- 2. Government CAP Integration: Connect with NDMA / SDMA Common Alerting Protocol for SMS early warnings.
- 3. In-situ IoT Telemetry: Integrate slope inclinometers, piezometers, and soil moisture sensors deployed along NH-6.
- 4. Full PWA Offline Capability: Offline map tile vector caching for disaster response teams in zero-connectivity areas.

14. GitHub Repository

Public Repository: https://github.com/shaik-irfa/slopegaurd_Ai

Branch: main

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